



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

TECHNICAL SCIENCES P2

MAY/JUNE 2024

MARKS: 75

TIME: 1½ hours

This question paper consists of 9 pages and 4 data sheets.

INSTRUCTIONS AND INFORMATION

1. Write your centre number and examination number in the appropriate spaces on the ANSWER BOOK.
2. This question paper consists of SIX questions. Answer ALL the questions in the ANSWER BOOK.
3. Start EACH question on a NEW page in the ANSWER BOOK.
4. Number the answers correctly according to the numbering system used in this question paper.
5. Leave ONE line between two subquestions, e.g. between QUESTION 2.1 and QUESTION 2.2.
6. You may use a non-programmable calculator.
7. You are advised to use the attached DATA SHEETS.
8. Round off your FINAL numerical answers to a minimum of TWO decimal places.
9. Give brief motivations, discussions, etc. where required.
10. Write neatly and legibly.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question numbers (1.1 to 1.5) in the ANSWER BOOK, e.g. 1.6 D.

- 1.1 The name of the functional group of ETHANOL is ...
A carboxyl.
B carbonyl.
C hydroxyl.
D formyl. (2)
- 1.2 Which homologous series has London forces, dipole-dipole forces and hydrogen bonds?
A Alkanes
B Ketones
C Aldehydes
D Carboxylic acids (2)
- 1.3 An element in group 3 that can be used for doping:
A Germanium
B Gallium
C Silicon
D Tin (2)
- 1.4 Which ONE of the following combinations of standard conditions is applicable to a galvanic cell with non-gaseous reactants and products?
A $1 \text{ mol} \cdot \text{dm}^{-3}$; 101,3 kPa; 25 K
B $1 \text{ mol} \cdot \text{dm}^{-3}$; 101,3 kPa
C $1 \text{ mol} \cdot \text{dm}^{-3}$; 298 K
D $1 \text{ mol} \cdot \text{dm}^{-3}$; 0 K (2)
- 1.5 A solution that can conduct an electric current through the movement of ions:
A Oxidising agent
B Reducing agent
C Electrolysis
D Electrolyte (2)

[10]

QUESTION 2 (Start on a new page.)

The table below represents six organic molecules with different functional groups.

A	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$	B	$\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \end{array}$
C	$\begin{array}{c} \text{H} \quad \quad \text{O} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$	D	C_5H_{12}
E	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$	F	C_3H_6

- 2.1 Define a *hydrocarbon*. (2)
- 2.2 Write down the letters that represent TWO unsaturated hydrocarbons. (2)
- 2.3 Write down the general formula of the following compounds:
- 2.3.1 **B** (1)
- 2.3.2 **F** (1)
- 2.4 Write down the IUPAC names of the following compounds:
- 2.4.1 **C** (2)
- 2.4.2 **D** (2)
- 2.5 Draw the structural formula of the following compounds:
- 2.5.1 **E** (2)
- 2.5.2 **F** (2)

[14]

QUESTION 3 (Start on a new page.)

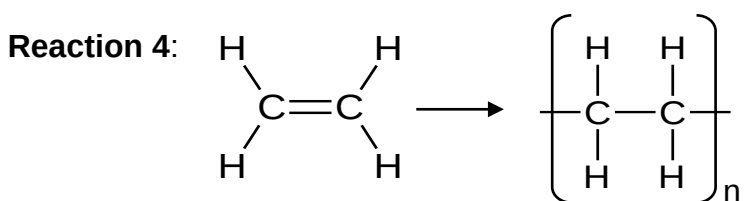
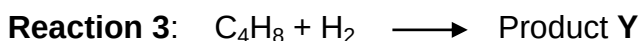
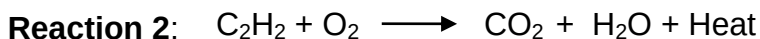
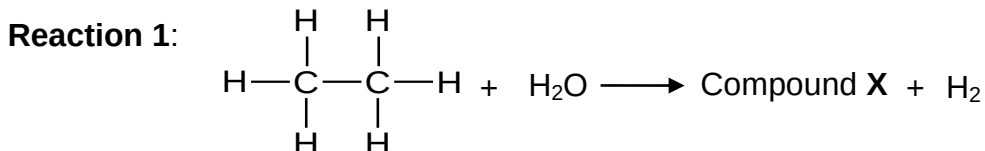
The following isomers are commonly used as fuels because of their low boiling points:

- A** Butane
B 2-methylpropane

- 3.1 Identify the homologous series to which these compounds belong. (1)
- 3.2 Define the term *structural isomers*. (2)
- 3.3 What type of isomerism is represented by the compounds above? (1)
- 3.4 Identify the type of intermolecular forces in both compounds. (1)
- 3.5 Which ONE of the compounds has the strongest intermolecular forces?
Write down only **A** or **B**. (1)
- 3.6 Give a reason for the answer to QUESTION 3.5. (2)
- 3.7 Define *vapour pressure*. (2)
- 3.8 Write down the compound above with a higher:
- 3.8.1 Boiling point (1)
- 3.8.2 Vapour pressure (1)
- [12]**

QUESTION 4 (Start on a new page.)

Consider the following organic reactions.



4.1 Consider reaction 1.

4.1.1 Name the type of reaction. (1)

4.1.2 Write down the CONDENSED STRUCTURAL FORMULA of compound X. (2)

4.1.3 Is compound X a PRIMARY, SECONDARY or TERTIARY alcohol? (1)

4.2 Consider reaction 2.

4.2.1 Balance the chemical equation for this reaction. (1)

4.2.2 Is this an EXOTHERMIC or ENDOTHERMIC reaction?
Give a reason. (3)

4.3 Consider reaction 3.

Product Y is used in cigarette lighters.

4.3.1 Name the type of ADDITION reaction taking place. (1)

4.3.2 Write down the IUPAC name of product Y. (2)

4.3.3 Write down the NAME or FORMULA of the catalyst used. (1)

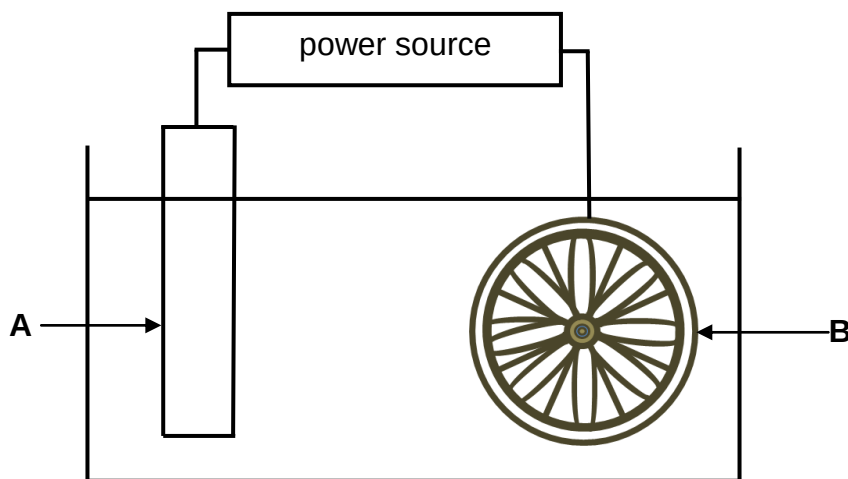
4.4 Reaction 4 is a polymerisation reaction used to produce plastics.

4.4.1 Define the term *polymer*. (2)

4.4.2 Write down the NAME of the monomer used in this reaction. (1)
[15]

QUESTION 5 (Start on a new page.)

Scratches on chromium mag wheels are removed by electroplating. The diagram below represents an electrolytic cell involved in such a process.

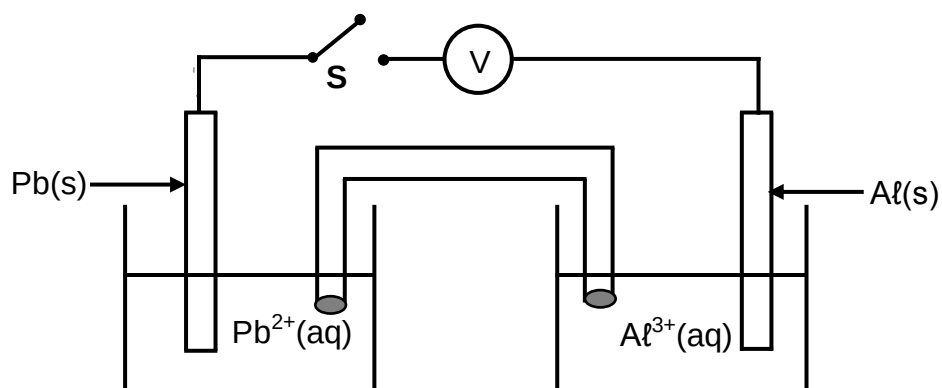


- 5.1 Define the term *electrolytic cell*. (2)
- 5.2 Write down the NAME or FORMULA of the metal used as the anode. (1)
- 5.3 Which electrode, **A** or **B**, is the cathode? Give a reason. (2)
- 5.4 Write down the half reaction taking place at the following electrodes:
- 5.4.1 **A** (2)
- 5.4.2 **B** (2)
- 5.5 What is the purpose of the power source? (1)
- 5.6 Why is the DC source preferred to the AC source? (2)

[12]

QUESTION 6 (Start on a new page.)

The diagram below represents an electrochemical cell operating under standard conditions.



- 6.1 What type of reaction is taking place in the cell above? (1)
- 6.2 What is the reading on the voltmeter? (1)
- 6.3 Switch **S** is now closed.
- 6.3.1 Calculate the reading on the voltmeter. (4)
- 6.3.2 Which ONE of the electrodes, **Pb** or **Al**, will experience a decrease in mass? Explain the answer. (3)
- 6.3.3 Write down the net ionic cell reaction. (3)
- [12]**

TOTAL: 75

**DATA FOR TECHNICAL SCIENCES GRADE 12
PAPER 2
GEGEWENS VIR TEGNIESE WETENSKAPPE GRAAD 12
VRAESTEL 2**

TABLE 1/TABEL 1: PHYSICAL CONSTANTS/FISIIESE KONSTANTES

NAME/NAAM	SYMBOL/SIMBOOL	VALUE/WAARDE
Standard pressure <i>Standaarddruk</i>	p^θ	$1,01 \times 10^5 \text{ Pa}$
Standard temperature <i>Standaardtemperatuur</i>	T^θ	$0^\circ\text{C}/273 \text{ K}$

TABLE 2/TABEL 2: FORMULAE/FORMULES

Emf/Emk	$E^\theta_{\text{cell}} = E^\theta_{\text{cathode}} - E^\theta_{\text{anode}} \quad / \quad E^\theta_{\text{sel}} = E^\theta_{\text{katode}} - E^\theta_{\text{anode}}$ <i>or/of</i> $E^\theta_{\text{cell}} = E^\theta_{\text{reduction}} - E^\theta_{\text{oxidation}} \quad / \quad E^\theta_{\text{sel}} = E^\theta_{\text{reduksie}} - E^\theta_{\text{oksidasie}}$ <i>or/of</i> $E^\theta_{\text{cell}} = E^\theta_{\text{oxidising agent}} - E^\theta_{\text{reducing agent}} \quad / \quad E^\theta_{\text{sel}} = E^\theta_{\text{oksideermiddel}} - E^\theta_{\text{reduseermiddel}}$
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Approximate relative atomic mass
Benaderde relatiewe atoommassa

Please turn over

TABLE 4A: STANDARD REDUCTION POTENTIALS
TABEL 4A: STANDAARD-REDUKSIEPOTENSIALE

Increasing strength of oxidising agents/Toenemende sterkte van oksideermiddels

Half-reactions/Halfreaksies		E° (V)
$\text{F}_2(\text{g}) + 2\text{e}^-$	$\rightleftharpoons 2\text{F}^-$	+ 2,87
$\text{Co}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Co}^{2+}$	+ 1,81
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons 2\text{H}_2\text{O}$	+1,77
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+ 1,51
$\text{Cl}_2(\text{g}) + 2\text{e}^-$	$\rightleftharpoons 2\text{Cl}^-$	+ 1,36
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^-$	$\rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+ 1,33
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$	$\rightleftharpoons 2\text{H}_2\text{O}$	+ 1,23
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{Mn}^{2+} + 2\text{H}_2\text{O}$	+ 1,23
$\text{Pt}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Pt}$	+ 1,20
$\text{Br}_2(\text{l}) + 2\text{e}^-$	$\rightleftharpoons 2\text{Br}^-$	+ 1,07
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons \text{NO}(\text{g}) + 2\text{H}_2\text{O}$	+ 0,96
$\text{Hg}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Hg}(\text{l})$	+ 0,85
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons \text{Ag}$	+ 0,80
$\text{NO}_3^- + 2\text{H}^+ + \text{e}^-$	$\rightleftharpoons \text{NO}_2(\text{g}) + \text{H}_2\text{O}$	+ 0,80
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Fe}^{2+}$	+ 0,77
$\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{H}_2\text{O}_2$	+ 0,68
$\text{I}_2 + 2\text{e}^-$	$\rightleftharpoons 2\text{I}^-$	+ 0,54
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons \text{Cu}$	+ 0,52
$\text{SO}_2 + 4\text{H}^+ + 4\text{e}^-$	$\rightleftharpoons \text{S} + 2\text{H}_2\text{O}$	+ 0,45
$2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$	$\rightleftharpoons 4\text{OH}^-$	+ 0,40
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cu}$	+ 0,34
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}$	+ 0,17
$\text{Cu}^{2+} + \text{e}^-$	$\rightleftharpoons \text{Cu}^+$	+ 0,16
$\text{Sn}^{4+} + 2\text{e}^-$	$\rightleftharpoons \text{Sn}^{2+}$	+ 0,15
$\text{S} + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{H}_2\text{S}(\text{g})$	+ 0,14
$2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{H}_2(\text{g})$	0,00
$\text{Fe}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Fe}$	- 0,06
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Pb}$	- 0,13
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Sn}$	- 0,14
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Ni}$	- 0,27
$\text{Co}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Co}$	- 0,28
$\text{Cd}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cd}$	- 0,40
$\text{Cr}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Cr}^{2+}$	- 0,41
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Fe}$	- 0,44
$\text{Cr}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Cr}$	- 0,74
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Zn}$	- 0,76
$2\text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons \text{H}_2(\text{g}) + 2\text{OH}^-$	- 0,83
$\text{Cr}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cr}$	- 0,91
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Mn}$	- 1,18
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Al}$	- 1,66
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Mg}$	- 2,36
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons \text{Na}$	- 2,71
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Ca}$	- 2,87
$\text{Sr}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Sr}$	- 2,89
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Ba}$	- 2,90
$\text{Cs}^+ + \text{e}^-$	$\rightleftharpoons \text{Cs}$	- 2,92
$\text{K}^+ + \text{e}^-$	$\rightleftharpoons \text{K}$	- 2,93
$\text{Li}^+ + \text{e}^-$	$\rightleftharpoons \text{Li}$	- 3,05

Increasing strength of reducing agents/Toenemende sterkte van reduceermiddels

TABLE 4B: STANDARD REDUCTION POTENTIALS
TABEL 4B: STANDAARD-REDUKSIEPOTENSIALE

Half-reactions/Halfreaksies		E° (V)
$\text{Li}^+ + \text{e}^-$	$\rightleftharpoons \text{Li}$	-3,05
$\text{K}^+ + \text{e}^-$	$\rightleftharpoons \text{K}$	-2,93
$\text{Cs}^+ + \text{e}^-$	$\rightleftharpoons \text{Cs}$	-2,92
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Ba}$	-2,90
$\text{Sr}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Sr}$	-2,89
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Ca}$	-2,87
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons \text{Na}$	-2,71
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Mg}$	-2,36
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Al}$	-1,66
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Mn}$	-1,18
$\text{Cr}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cr}$	-0,91
$2\text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons \text{H}_2(\text{g}) + 2\text{OH}^-$	-0,83
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Zn}$	-0,76
$\text{Cr}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Cr}$	-0,74
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Fe}$	-0,44
$\text{Cr}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Cr}^{2+}$	-0,41
$\text{Cd}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cd}$	-0,40
$\text{Co}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Co}$	-0,28
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Ni}$	-0,27
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Sn}$	-0,14
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Pb}$	-0,13
$\text{Fe}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Fe}$	-0,06
$2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{H}_2(\text{g})$	0,00
$\text{S} + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{H}_2\text{S}(\text{g})$	+0,14
$\text{Sn}^{4+} + 2\text{e}^-$	$\rightleftharpoons \text{Sn}^{2+}$	+0,15
$\text{Cu}^{2+} + \text{e}^-$	$\rightleftharpoons \text{Cu}^+$	+0,16
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}$	+0,17
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cu}$	+0,34
$2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$	$\rightleftharpoons 4\text{OH}^-$	+0,40
$\text{SO}_2 + 4\text{H}^+ + 4\text{e}^-$	$\rightleftharpoons \text{S} + 2\text{H}_2\text{O}$	+0,45
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons \text{Cu}$	+0,52
$\text{I}_2 + 2\text{e}^-$	$\rightleftharpoons 2\text{I}^-$	+0,54
$\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{H}_2\text{O}_2$	+0,68
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Fe}^{2+}$	+0,77
$\text{NO}_3^- + 2\text{H}^+ + \text{e}^-$	$\rightleftharpoons \text{NO}_2(\text{g}) + \text{H}_2\text{O}$	+0,80
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons \text{Ag}$	+0,80
$\text{Hg}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Hg}(\ell)$	+0,85
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons \text{NO}(\text{g}) + 2\text{H}_2\text{O}$	+0,96
$\text{Br}_2(\ell) + 2\text{e}^-$	$\rightleftharpoons 2\text{Br}^-$	+1,07
$\text{Pt}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Pt}$	+1,20
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons \text{Mn}^{2+} + 2\text{H}_2\text{O}$	+1,23
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$	$\rightleftharpoons 2\text{H}_2\text{O}$	+1,23
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^-$	$\rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+1,33
$\text{Cl}_2(\text{g}) + 2\text{e}^-$	$\rightleftharpoons 2\text{Cl}^-$	+1,36
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+1,51
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons 2\text{H}_2\text{O}$	+1,77
$\text{Co}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Co}^{2+}$	+1,81
$\text{F}_2(\text{g}) + 2\text{e}^-$	$\rightleftharpoons 2\text{F}^-$	+2,87

Increasing strength of oxidising agents/Toenemende sterkte van oksideermiddels

Increasing strength of reducing agents/Toenemende sterkte van reduceermiddels